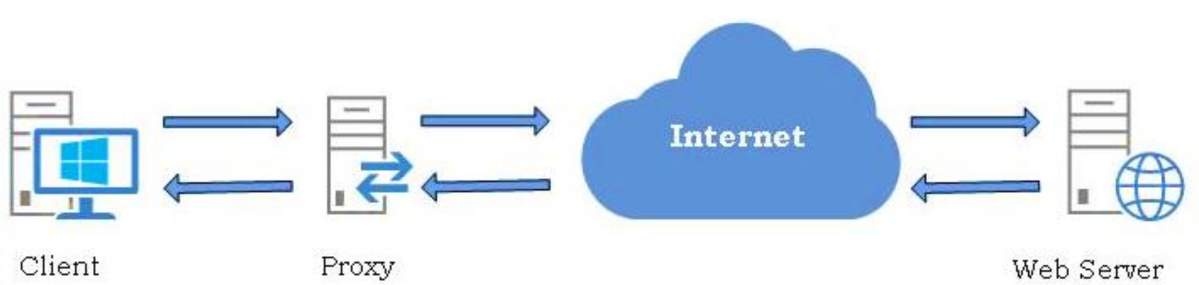
Basic Terminologies:

Proxy Server:

- The term 'Proxy' is a contraction that comes from the middle English word procuracy.
- Proxy meaning to act on behalf of another, sits between two entities & performs service.
- In networking and web traffic, proxy is a device or server that acts on behalf of other.
- It is serves as a mediator for requests from clients asking resources from other servers.
- There no direct communication occurs between the client and the destination server.
- Proxy Server takes the requests from a client, puts that client on hold for a moment.
- Makes the requests as if it is its own request out to the final destination servers.
- Proxy Servers are memory & disk intensive & single point of failure in the network.
- Proxy server or application-level gateway acts as a gateway between client & internet.
- Proxy server acts as an intermediary between your devices and the internet as a whole.
- A proxy server basically creates a gateway between you as client and the internet.

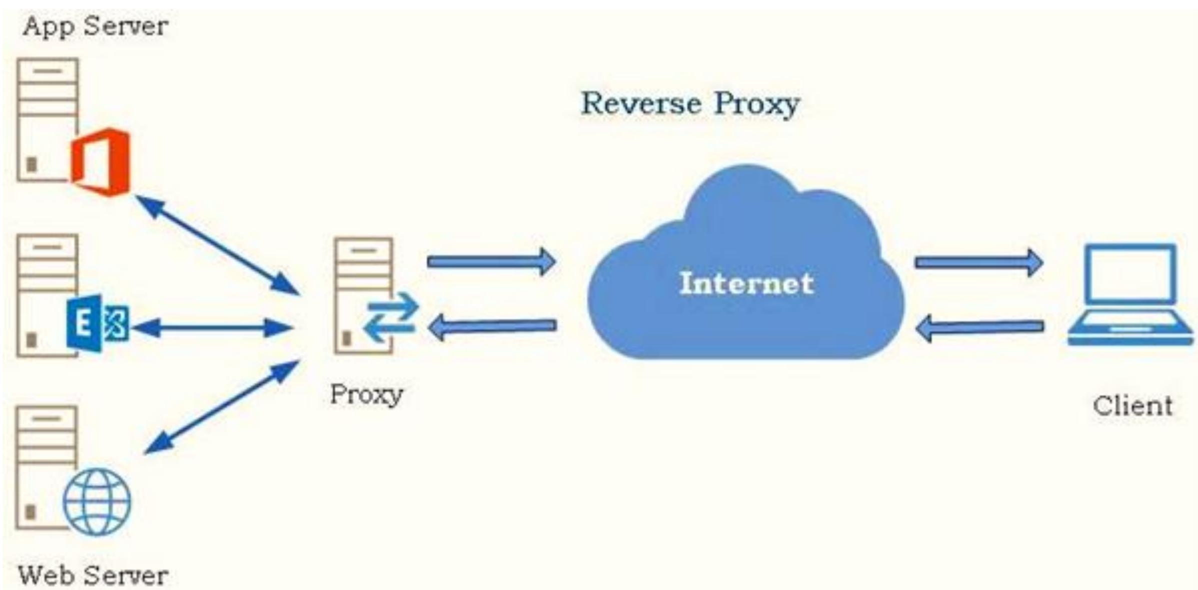
Forward Proxy:

- Takes origin connections from intranet clients & connect to servers outside on internet.
- Forward proxy takes requests from an internal LAN network & forwards them Internet.
- Sometimes, forward proxy may even serve the requesting client with cached information.
- When end user web requests are forwarded to a proxy before going out to the internet.
- And responses go back through the proxy and then back to the user for privacy & control.
- Main purpose of forward proxy server is to help users access the servers over the internet.
- Forward Proxy Server is the most popular proxy mode and it present in almost all networks.
- FP, which forward the request from the intranet clients (browser) to the internet servers.



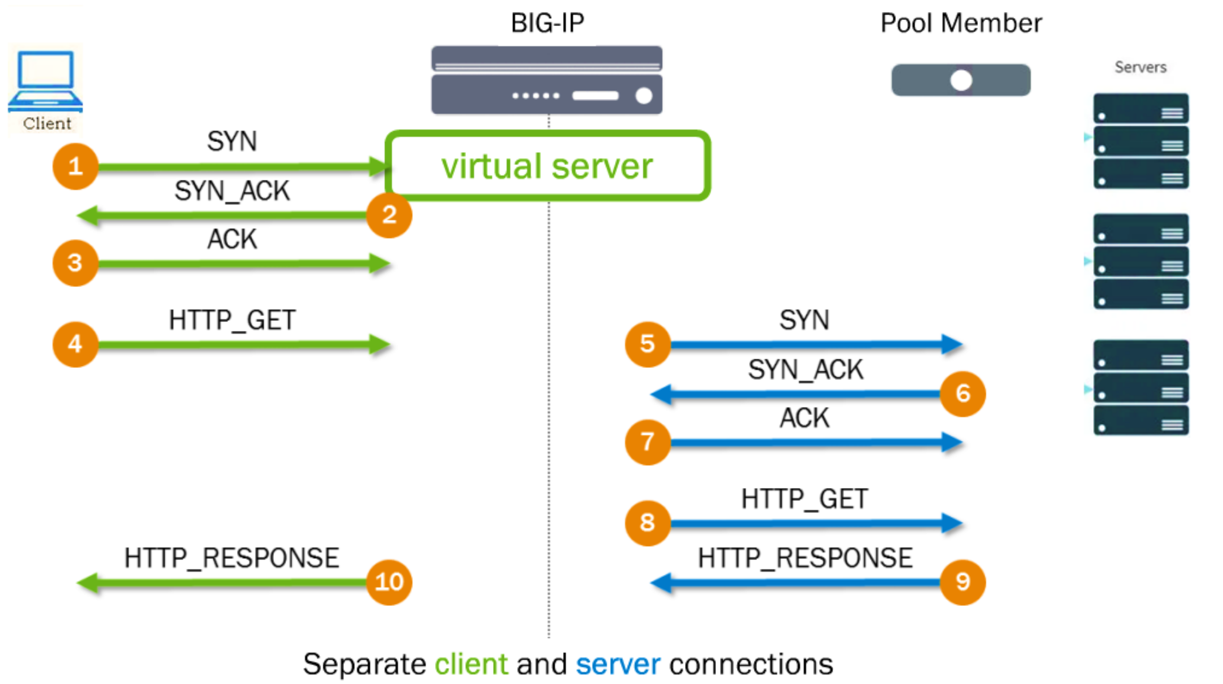
Reverse Proxy:

- A Proxy Server which takes requests from external clients (web browsers) or Internet.
- And forwards them to servers in an internal network is called as Reverse Proxy Server.
- Acts as a front-end server accept the requests and forward to Application servers web.
- Reverse proxy servers & load balancers are components in a client-server architecture.
- Both act as intermediaries in the communication between the clients and the servers.
- A reverse proxy accepts a request from a client, forwards it to a server that can fulfill it.
- And a reverse proxy returns the server’s response to the requested internet client back.
- With a reverse proxy the clients believe you are contacting the external server directly.
- In reality, contacting the reverse proxy server, that pretends to be the external server.
- The reverse proxy will then make another request to the real server on the client behalf.



Full Proxy:

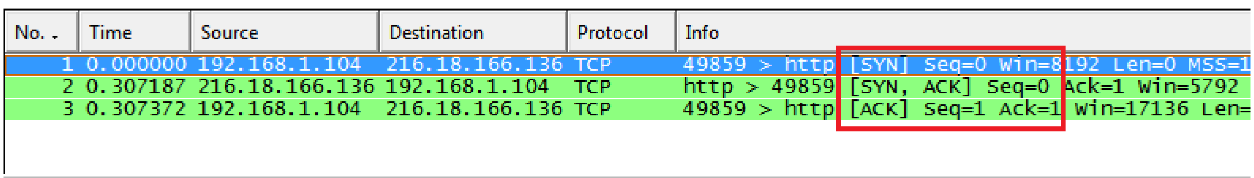
- Full proxy creates client connection along with separate server connection with little gap.
- Client connects to proxy on one end & proxy establishes separate, connection to the server.
- A full Proxy establishes separate connection, means this is bi-directionally on both sides.
- Maintains two separate connections, one between the proxy server device and the client.
- Another connection one between the proxy server device and the destination server.
- In Full proxy there is never any blending of connections from the client side to server side.
- Full proxy can manipulate, inspect, drop, do what need to traffic on both sides & directions.
- When clients make request from internet, terminate on reverse proxy sitting front of server.
- Reverse proxies are good for traditional load balancing, optimization and SSL offloading etc.
- Full proxies are named because of they completely proxy connections incoming & outgoing.



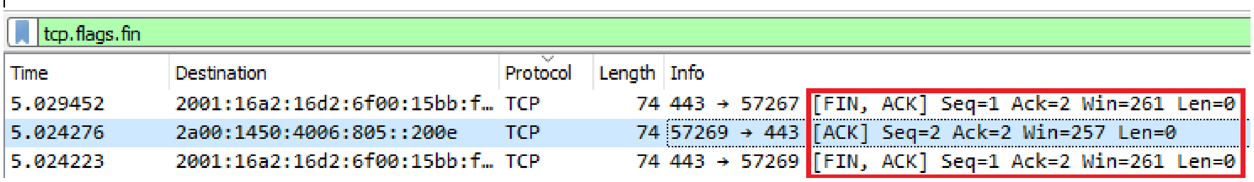
TCP 3-Way Handshake:

- 3-Way Handshake is a process which is used in a TCP/IP network to make connection.
- It is 3-step process that requires both the client & server to exchange synchronization.
- And acknowledgment packets before the real data communication process starts.
- TCP is a connection-oriented protocol, connection needs to be established before.
- Connection needs to be established before two devices can communicate each other.
- TCP uses three-way handshake to negotiate sequence and acknowledgment fields.
- Allows to transfer multiple TCP socket connections in both directions at the same time.

Message	Description
Syn	Used to initiate and establish a connection. It also helps you to synchronize sequence numbers between devices.
ACK	Helps to confirm to the other side that it has received the SYN.
SYN-ACK	SYN message from local device and ACK of the earlier packet.
FIN	Used to terminate a connection.

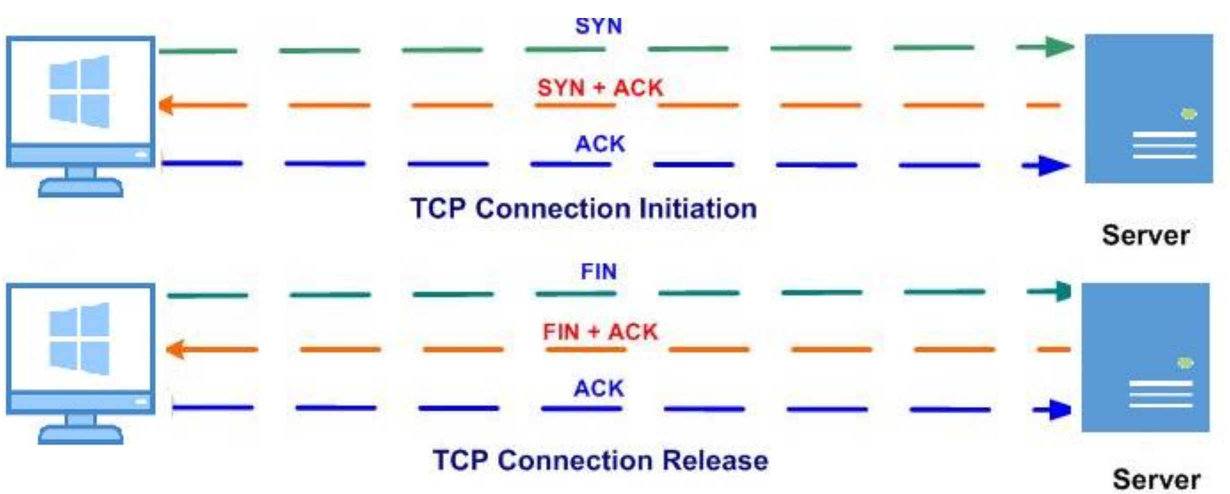


No.	Time	Source	Destination	Protocol	Info
1	0.000000	192.168.1.104	216.18.166.136	TCP	49859 > http [SYN] Seq=0 Win=8192 Len=0 MSS=1
2	0.307187	216.18.166.136	192.168.1.104	TCP	http > 49859 [SYN, ACK] Seq=0 Ack=1 Win=5792
3	0.307372	192.168.1.104	216.18.166.136	TCP	49859 > http [ACK] Seq=1 Ack=1 Win=17136 Len=



tcp.flags.fin

Time	Destination	Protocol	Length	Info
5.029452	2001:16a2:16d2:6f00:15bb:f…	TCP	74	443 → 57267 [FIN, ACK] Seq=1 Ack=2 Win=261 Len=0
5.024276	2a00:1450:4006:805::200e	TCP	74	57269 → 443 [ACK] Seq=2 Ack=2 Win=257 Len=0
5.024223	2001:16a2:16d2:6f00:15bb:f…	TCP	74	443 → 57269 [FIN, ACK] Seq=1 Ack=2 Win=261 Len=0

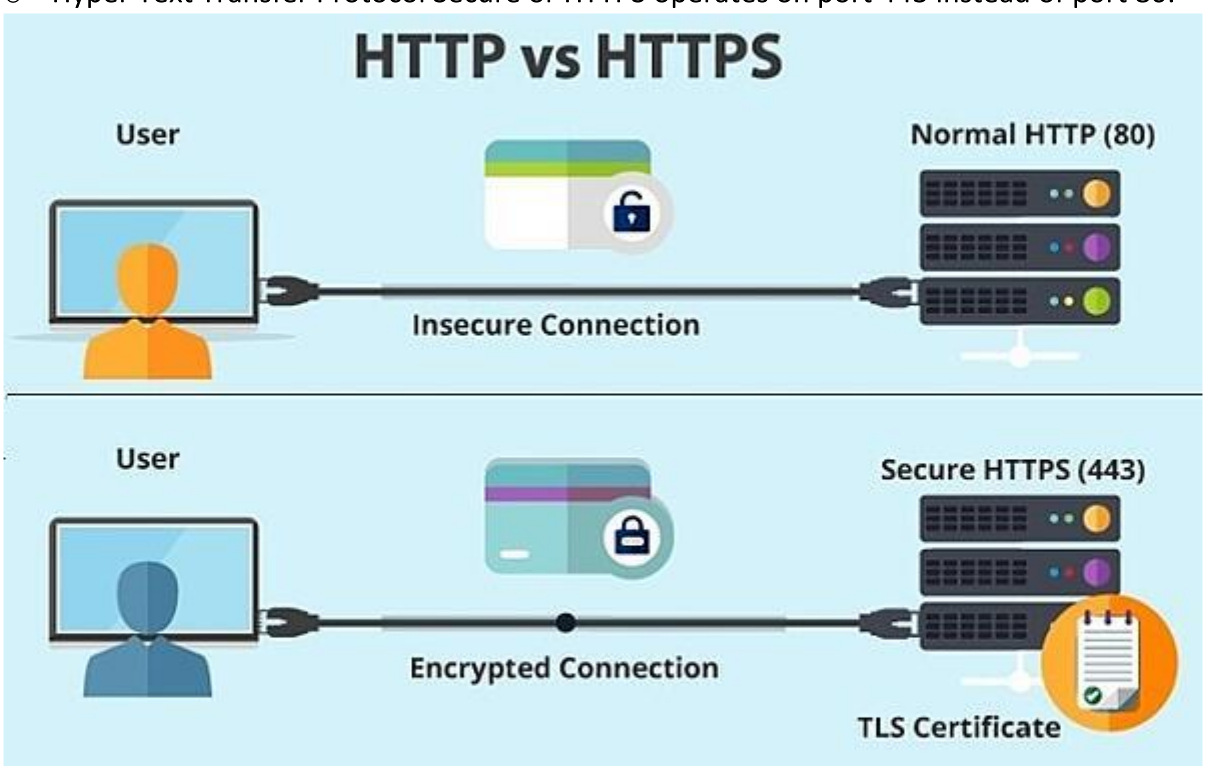


HTTP:

- Hypertext Transfer Protocol is designed to enable communication between client & server.
- Hypertext Transfer Protocol works as request-response protocol between client and server.
- Client sends an HTTP request to the server; then the server returns a response to the client.
- Response contains status information about request & may also contain requested content.
- Hypertext Transfer Protocol (HTTP) request is the way internet communications platforms.
- Such as web browsers in any computer ask for the information, they need to load a website.
- two of the most common Hypertext Transfer Protocol HTTP methods are ‘GET’ and ‘POST’;
- The GET request expects information back in return (Usually in the form of a Website).
- While POST request typically indicates that client is submitting information to web server.
- HTTP headers contain text info store in key-value pairs & are include in every HTTP request.
- HTTP header contain text info store in key-value pairs & are include in every HTTP response.
- Headers communicate core info, such as what browser client is using data is being request.

HTTPS (Secure HTTP):

- Hyper Text Transfer Protocol Secure or HTTPS is just like HTTP except it is used SSL/TLS.
- HTTPS together with Secure Sockets Layer (SSL) and Transport Layer Security (TLS).
- It provides a security layer on top of the Hypertext Transfer Protocol (HTTP) protocol.
- Encrypts data, ensures identity of both devices & makes sure, data has not been modified.
- The HTTPS is a standard that is highly used by online banking and online shopping etc.
- When you use HTTPS, instead of the original http:// it uses the https:// prefix in URLs.
- Hyper Text Transfer Protocol Secure or HTTPS operates on port 443 instead of port 80.



TCP and UDP:

TCP is the abbreviation of " Transmission Control Protocol" whereas UDP is the abbreviation of "User Datagram Protocol". TCP and UDP are both the main protocols which are used during the Transport layer of a TCP/IP Model. Both of these protocols are involved in the process of transmission of data. While UDP is used in situations where the volume of data is large and security of data is not of much significance, TCP is used in those situations where security of data is one of the main issues. Both TCP and UDP are protocols used for sending bits of data known as packets over the Internet. They both build on top of the Internet protocol. In other words, whether you are sending a packet via TCP or UDP, that packet is sent to an IP address. TCP and UDP are not the only protocols that work on top of IP. However, they are the most widely used. The widely used term “TCP/IP” refers to TCP over IP.

TCP and UDP Ports:

As you know every computer or device on the Internet must have a unique number assigned to it called the IP address. This IP address is used to recognize your particular computer out of the millions of other computers connected to the Internet. When information is sent over the Internet to your computer . It accepts that information by using TCP or UDP ports.

An easy way to understand ports is to imagine your IP address is a cable box and the ports are the different channels on that cable box. The cable company knows how to send cable to your cable box based upon a unique serial number associated with that box (IP Address), and then you receive the individual shows on different channels (Ports).

Ports work the same way. You have an IP address, & then many ports on that IP address. You can have a total of 65,535 TCP Ports and another 65,535 UDP ports. When a program on your computer sends or receives data over the Internet it sends that data to an ip address and a specific port on the remote computer and receives the data on a usually random port on its own computer. If it uses the TCP protocol to send and receive the data, then it will connect & bind itself to a TCP port. If it uses UDP protocol to send and receive data, it will use a UDP port.

<-------------------- 192.168.1.10 -------------------->

0	1	2	3	4	5	..	..	..	..	..	..	..	..	65531	65532	65533	65534	65535

IP address with Ports

Port Range Groups	Description
0 to 1023	Well Known Port Numbers
1024 to 49151	Registered Ports
49152 to 65536	Dynamic or Private Ports or Public Ports

Port Number	Usage	Port Number	Usage	Port Number	Usage
20 & 21	FTP	23	Telnet	443	HTTPS, SSL/TLS
22	SSH	25	SMTP	161	SNMP
53	DNS	80	HTTP	123	NTP